

RAGHAVENDRA GOLLA

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PROFESSIONAL SUMMARY

Data Scientist & Applied ML Researcher with a strong mathematical foundation (BSc Mathematics) and pursuing MSc Data Science. Hands-on experience developing end-to-end machine learning pipelines, deep learning architectures (PyTorch CNNs), NLP text classifiers, and interactive business intelligence dashboards (Power BI, Streamlit). Proficient in Python, SQL, Scikit-Learn, and Explainable AI (SHAP/LIME). Passionate about transforming complex datasets into actionable business intelligence and scalable predictive systems.

TECHNICAL SKILLS

Programming & Databases: Python (Pandas, NumPy, Scikit-Learn, PyTorch), SQL (MySQL, PostgreSQL), MongoDB

Data Analytics & BI: Power BI, Advanced Excel, Exploratory Data Analysis (EDA), Matplotlib, Seaborn, Plotly

Machine Learning & AI: Supervised/Unsupervised ML, Deep Learning (CNNs), NLP (TF-IDF), Explainable AI (SHAP, LIME)

Web & Deployment Tools: Streamlit, FastAPI, Git, GitHub, Linux/Bash, Jupyter Notebooks

Core Competencies: Feature Engineering, Statistical Hypothesis Testing, Predictive Modeling, Model Interpretability

PROFESSIONAL EXPERIENCE & INTERNSHIPS

Machine Learning Intern | *SmartED Innovations*

Feb 2026 – March 2026

- Engineered end-to-end machine learning workflows using Python, Pandas, and Scikit-Learn.
- Performed comprehensive data preprocessing, outlier detection, and feature encoding across tabular datasets.
- Benchmarked multiple supervised classification and regression algorithms, optimizing hyperparameters to maximize cross-validation performance.
- Documented model evaluation metrics (Precision, Recall, F1-Score, RMSE) and prepared technical reports for stakeholders.

FEATURED PROJECTS

Explainable ML for Cardiovascular Disease Risk | *Python, Scikit-Learn, SHAP, LIME*

- Implemented interpretable predictive models to forecast early cardiovascular disease risk.
- Integrated SHAP (Shapley Additive exPlanations) and LIME to generate local feature attribution plots and global feature importance rankings for clinical transparency.
- Formatted findings into an IEEE-standard academic research paper on interpretable healthcare AI.

Handwritten Digit Recognition System | *PyTorch, CNN, Computer Vision, Streamlit*

- Built a deep Convolutional Neural Network (CNN) architecture with batch normalization, dropout, and pooling layers trained on the MNIST dataset.
- Achieved ~99% test classification accuracy with robust generalization.
- Deployed an interactive Streamlit web application enabling real-time drawing canvas inference and live class probability charts.

Social Media Depression Text Classifier | *NLP, Scikit-Learn, TF-IDF, Naive Bayes*

- Developed a natural language processing text classification pipeline to detect psychological distress markers in social media posts.
- Extracted 10,000 TF-IDF n-gram features and benchmarked Multinomial Naive Bayes, Logistic Regression, and Linear SVMs.
- Multinomial Naive Bayes achieved top performance with a 95.7% ROC-AUC score.

California Housing Price Estimator | *Python, Scikit-Learn, Streamlit*

- Built an ensemble regression pipeline on 20,640 records across 8 economic and geographical features.
- Implemented automated preprocessing pipelines with StandardScaler and handled multicollinearity.
- Deployed a responsive Streamlit web application for real-time house price estimation.

EDUCATION

Master of Science (MSc) in Data Science | *Chandigarh University, India*

2025 – 2027

- Specialization: Machine Learning, Statistical Inference, Deep Learning, Big Data Analytics.

Bachelor of Science (BSc) in Mathematics | *Krishna University, Andhra Pradesh, India*

2017 – 2020

- Core coursework: Linear Algebra, Differential Calculus, Probability Theory, Mathematical Statistics.

CERTIFICATIONS

IBM Data Analyst Professional Certificate

- IBM / Coursera (Comprehensive SQL, Python, Excel & Power BI)

Machine Learning Certification

- SmartED Innovations (2026)